EBU6018

Group 2 Class Test Question Sheet **16th December 2022**

OPEN BOOK, CALCULATORS ALLOWED.

Please read the following instructions carefully:

- **Type your answers in the supplied answer sheet; hand-written equations or sketches can be incorporated into the answer sheet**
- **Submit the ANSWER SHEET as a PDF file to QMplus.**
- **For calculations, you must show all the working. If you do not you will get 0 marks for the question.**
- **Multiple submissions with identical solutions will receive 0 marks.**
- **If you require any assistance, raise the issue via the EBU6018 student forum on QMplus, immediately.**

Question 1.

This question is about the Karhunen-Loeve Transform (KLT).

Table 1 shows a 2D data set which contains a sample of a larger set.

x	y
-0.98	1.27
1.11	0.03
-3.21	-2.39
-1.31	5.01

Table 1

Calculate the covariance matrix to 4 decimal places. **[15 marks]**

Question 2.

This question is about the Discrete Cosine Transform (DCT).

i) Determine the DCT output sequence y[n] (to 4 decimal places) if the input sequence is:

$$s[n] = [1.9, 1.2, 0.9, -0.9, -0.7, -0.8, 0.1, 1.6]$$

[10 marks]

ii) Then, briefly discuss what information about the input s[n] can be obtained from the output y[n]?

[5 marks]